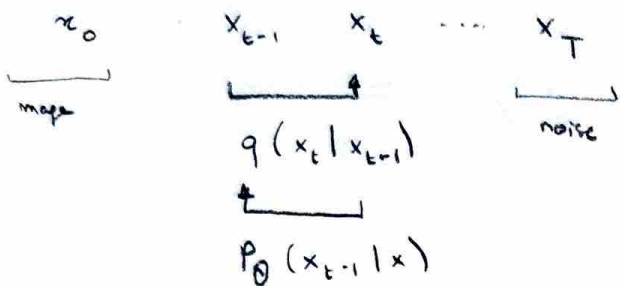


Problem Formulation

- we have a distribution, we want to sample a new sample from that distribution.
- start from noise $\longrightarrow$ diversity

Forward and Backward Process



Forward Process: adding noise.

Reverse Process: learn to denoise.

Probability Distribution in high dimensional space: $x \sim N(\mu, \Sigma)$

$$\begin{pmatrix} x^{(1)} \\ \vdots \\ x^{(n)} \end{pmatrix} \sim N \left(\begin{pmatrix} \mu^1 \\ \vdots \\ \mu^n \end{pmatrix}, \begin{pmatrix} \Sigma^{11} & \Sigma^{1n} \\ \Sigma^{n1} & \Sigma^{nn} \end{pmatrix} \right)$$

usual assumption: $\Sigma = \sigma^2 I = \begin{pmatrix} \sigma^2 & & 0 \\ & \ddots & \\ 0 & & \sigma^2 \end{pmatrix}$

isotropic distribution: variance is the same across all dimensions.



Gaussian probability density: $p(x) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp \left(-\frac{\|x - \mu\|^2}{2\sigma^2} \right)$

Noise schedule: weight $\times$ image + weight $\times$ noise

$$q(x_t | x_{t-1}) = N(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

$$\longrightarrow x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon \quad \text{where } \epsilon \sim N(0, 1)$$

- start with little noise and gradually increase the amount of noise added.
this is because when you are close to the real image, you want to learn the fine grained details.

$$0 \leq \beta_1 < \beta_2 < \dots < \beta_T \leq 1$$

• there is a way to get all maps in the nonrec schedule (not sequentially). ^②

$$z_t = \sqrt{1-\beta_t} z_{t-1} + \sqrt{\beta_t} \epsilon$$

$$z_{t-1} = \sqrt{1-\beta_{t-1}} z_{t-2} + \sqrt{\beta_{t-1}} \epsilon$$

$$\begin{aligned} z_t &= \sqrt{1-\beta_t} (\sqrt{1-\beta_{t-1}} z_{t-2} + \sqrt{\beta_{t-1}} \epsilon) + \sqrt{\beta_t} \epsilon \\ &= \sqrt{1-\beta_t} \sqrt{1-\beta_{t-1}} z_{t-2} + (\sqrt{1-\beta_t} \sqrt{\beta_{t-1}} + \sqrt{\beta_t}) \epsilon \end{aligned}$$

reparameterization

$$z_t = \sqrt{\alpha_t} z_{t-1} + \sqrt{1-\alpha_t} \epsilon$$

$$z_{t-1} = \sqrt{\alpha_{t-1}} z_{t-2} + \sqrt{1-\alpha_{t-1}} \epsilon$$

$$z_t = \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} z_{t-2} + \sqrt{\alpha_t} \sqrt{1-\alpha_{t-1}} \epsilon + \sqrt{1-\alpha_t} \epsilon$$

Gaussian distribution property: if you have two variables that are independent and [...]

* * variance of sum of independent Gaussians is sum of respective variances.

$$\text{variance 1: } (\alpha_t)(1-\alpha_{t-1}) \quad \text{variance 2: } (1-\alpha_t)$$

$$\text{sum: } \alpha_t - \alpha_t \alpha_{t-1} + 1 - \alpha_t = 1 - \alpha_t \alpha_{t-1}$$

$$\rightarrow z_t = \sqrt{\alpha_t \alpha_{t-1}} z_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \epsilon \quad \epsilon \sim N(0, 1)$$

$$\text{objective: } \max_{\theta} \log p_{\theta}(x_0)$$

Joint Probability Distribution: $p(x_1, x_2) = p(x_1) \times p(x_2 | x_1)$

heatmaps are useful to represent joint probability distributions -

Marginalization: $p(x_1) = \int p(x_1, x_2) dx_2$ sum across all x_2 .

$$p(x_1, x_2, \dots, x_T) = p(x_1) \times p(x_2 | x_1) \times \dots \times p(x_T | x_1, \dots, x_{T-1})$$

$$= p(x_{1:T})$$

Marginal Dichtung

$$p(x_1) = \int p(x_{1:T}) dx_{2:T}$$

Problem: $\log p_0(x_0)$

the way to obtain x_0 is sampling a noise distribution.

take joint probability distribution over all timesteps. + sum across all different quantities except x_0 .
+ aka marginalize.

$$\log p_{\theta}(x_0) = \log \int p_{\theta}(x_0, x_{1:T}) dx_{1:T}$$

$\hookrightarrow$ this is not computable.

• in diffusion models, $\log p_\theta(x_0)$ is the log likelihood the model assigns to a clean data point.

- This is not possible to compute because it requires maximization over all possible trading trajectories.

$$x_0 \xrightarrow{\text{clear}} x_1 \xrightarrow{\text{noise}} \dots \xrightarrow{\text{noise}} x_T$$

to obtain y_0 we start with noise and denoise at each step -

$$\bullet \log p_{\theta}(x_0) = \log \int p_{\theta}(x_0, y_{1:T}) dx_{1:T}$$

Here joint means, $p(x_0) = \int p(x_0, z) dz$ where z is a latent we integrate in order to compute $p(x_0)$.

• In the case of diffusion, we have

$$p_{\theta}(x_0) = \int p(x_0, x_1) dx_1 = \int p(x_0 | x_1) p(x_1) dx_1 = \int p(x_0 | x_1) \left[\int p(x_1 | x_2) p(x_2) dx_2 \right] dx_1$$

$$p_0(x_1) = \int p(x_1, x_2) dx_2 = \int p(x_1 | x_2) p(x_2) dx_2$$

$$p_{\theta}(x_0) = \int \dots \int p(x_0 | x_1) \dots p(x_{T-1} | x_T) p(x_T) dx_1 dx_2 \dots dx_T.$$

$$p_\theta(x_0) = \int p_\theta(x_0, x_1) dx_1 = \int p_\theta(x_0 | x_1) p(x_1) dx_1$$

$$p_\theta(x_1) = \int p_\theta(x_1, x_2) dx_2 = \int p_\theta(x_1 | x_2) p(x_2) dx_2$$

$$p_\theta(x_0) = \int p_\theta(x_0 | x_1) \int p_\theta(x_1 | x_2) p(x_2) dx_2$$

$$= \int \int p_\theta(x_0 | x_1) p_\theta(x_1 | x_2) p(x_2) dx_1 dx_2$$

$$= \int \int \dots \int p_\theta(x_0 | x_1) p(x_1 | x_2) \dots p(x_{T-1} | x_T) p(x_T) dx_1 dx_2 \dots dx_T$$

$$= \int \dots \int p_\theta(x_0, x_1, \dots, x_T) dx_1 \dots dx_T \Rightarrow \text{joint distribution}$$

Derivation of the tractable loss

↳ Markov Assumption:

$$p(x_{t-1} | x_t, x_{t+1}, \dots, x_T) = p(x_{t-1} | x_t)$$

because to compute x_{t-1} , you only need x_t .

Trick: we have a forward process that we defined

$$p_\theta(x_0) = \int p_\theta(x_0, x_{1:T}) dx_{1:T}$$

$$= \int \frac{p_\theta(x_{0:T}) \times q(x_{1:T} | x_0)}{q(x_{1:T} | x_0)} dx_{1:T}$$

$$= \mathbb{E}_{x_{1:T} \sim q(x_{1:T} | x_0)} \left[\frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right]$$

$$\log \left(\mathbb{E} \left[\frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right] \right) \geq \mathbb{E} \left[\log \left(\frac{p_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right) \right]$$

↳ Jensen's Inequality.



STEP 1: Derivation of a Lower Bound

KL Divergence

STEP 2: Expand terms of the lower bound

(5)

$$KL(P \parallel Q) = \sum_{i=1}^n p_i \log\left(\frac{p_i}{q_i}\right)$$

quantifies the distance between distributions.

continuous case: $KL(P \parallel Q) = \int \log\left(\frac{p(x)}{q(x)}\right) p(x) dx = \mathbb{E}_{x \sim p} \left[\log\left(\frac{p(x)}{q(x)}\right) \right]$

$$ELBO = - \sum_{t=2}^T KL(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t)) + \text{some other terms}$$

can we compute this?
 we want to learn θ .

STEP 3: Show lower bound is tractable (double check)

(a) show $q(x_{t-1} | x_t, x_0)$ is tractable

what is the distribution of less noisy image given a noisy image & clean image

Bayes Rule: $P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$

$$q(x_{t-1} | x_t, x_0) = \frac{q(x_t | x_{t-1}, x_0) q(x_{t-1} | x_0)}{q(x_t | x_0)}$$

Bayes Rule.

Markov Assumption:

$$q(x_t | x_{0:t-1}) = q(x_t | x_{t-1}) \Rightarrow q(x_t | x_{t-1}, x_0) = q(x_t | x_{t-1})$$

$$\begin{aligned} q(x_{t-1} | x_t, x_0) &\propto q(x_t | x_{t-1}) q(x_{t-1} | x_0) \\ &= N(\tilde{\mu}_t, \tilde{\Sigma}_t) \end{aligned}$$

(b) show $p_\theta(x_{t-1} | x_t)$ is tractable: Assume p_θ is Gaussian

$$p_\theta(x_{t-1} | x_t) = N(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Note:

$q(x_t | x_{t-1})$ is gaussian

$\rightarrow q(x_{t-1} | x_t)$ is approx Gaussian

STEP 4: Deduce final Loss Function

(6)

$$\mathcal{L}_{DDPM} = \mathbb{E}_{t, x_0, \epsilon} [\| \epsilon_{\theta}(\sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \epsilon, t) - \epsilon \|^2]$$

$$t \sim \mathcal{U}\{1, T\} \quad x_0 \sim q_0(x_0) \quad \epsilon \sim N(0, I)$$

- L2 regression on the noise from clear image to the noisy image
- expectation over
 - t : noise level
 - x_0 : framy set
 - ϵ : random noise
- model's prediction of the noise you added.
- the model can understand at which step of denoising it is since we give t .

Training Recipe:

- ① sample clear image sample noise sample timestep
 $x_0 \sim q_0(x_0)$ $\epsilon \sim N(0, I)$ $t \sim \mathcal{U}\{1, T\}$

$$\text{Noised image: } x_t = \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \epsilon$$

- ② use x_t and t to predict ϵ via $\epsilon_{\theta}(x_t, t)$.
predict the noise added using model.

Inference Recipe ① sample noise $x_T \sim N(0, I)$

- ② perform iterative update x_t to x_{t-1} .

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

- ③ Obtain x_0 .

Limitations

T is usually large $\rightarrow$ 1000+ in the original paper

* reducing inference costs

Attempts: derive the formula by induction & step steps.